

6. MASS AND BALANCE

6.1 AUXILIARY FUEL TANK LOADING DATA

Auxiliary Fuel Tank Max Quantity	Mass (kg)	Arm (mm)	Massmoment (kgmm)
200 Liter (Single tank)	160.0	3800	608000
400 Liter (Both tanks)	320.0	3800	1216000
NOTE Fuel mass values are based upon a fuel density of 0.8 kg/l.			

6.2 CALCULATING LONGITUDINAL CG

Follow the loading examples shown in the basic FLM, Section 6 and add the plot of the aux tank fuel data.

NOTE To make certain that the CG does not wander outside of the CG envelope during the flight, the helicopter must be so loaded that the progressive plots of the minimum landing mass, the supply tank fuel, the main tank fuel, and the auxiliary tank fuel all lay within the approved CG limits and that the arm of the last plot is equal to or less than 3375 mm.

6.3 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Long Range Fuel System				
– Installation Provisions	105-82101	6.2	3800	23560
– Installation Provisions	105-82102	5.0	3800	19000
– Installation Provisions	105-82103	4.6	3800	17480
– Installation Provisions	D133-1554	6.2	3800	23560
– Auxiliary Fuel Tank-LH	105-82201	24.5	3800	93100
– Auxiliary Fuel Tank-RH	105-82206	24.5	3800	93100
– Auxiliary Fuel Tank	D133-1207	24.5	3800	93100
– Tank Protective Panel	105-82166	1.1	3800	4180
– Tank Protective Panel	D133-1605	1.1	3800	4180